Computational Economics
https://doi.org/10.1007/s10614-025-11127-4

TransGuard-QNet: A Hybrid Quantum Machine Learning Framework for Financial Transaction Risk Assessment and Fraud Detection With Linear Programming Optimization

T. S. Sasikala[1] · S. Sivakami[1] · Prem Nisha G.[2] · R. Saranya[3]

Received: 29 January 2025 / Accepted: 18 September 2025
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract
The patterns of fraud across transactions have never been more complex, creating a challenge to financial institutions in terms of assessing and reducing risk. An integrated hybrid Quantum Machine Learning (QML) framework, TransGuard-QNet, is presented in this research for financial transaction risk assessment and prediction, which combines supervised and unsupervised models with Linear Programming (LP) for decision optimization. After preprocessing the input features with several preprocessing techniques, the proposed model oversees outliers via Isolation Forest (iForest) for improved robustness. A feature engineering phase primarily relies on mathematics, behaviors, statistics, frequency, and correlation features. Besides, it uses an Improved Extreme Gradient Boosting (iXGBoost) and Quantum Neural Nets (QNNs) model to predict risk probability, and Quantum Autoencoders (QAEs) and k-Means clustering techniques to find anomalies. The resulting predictions are combined to form a final output based on the optimized parameters with a focus on achieving both high predictive accuracy and anomaly detection effectiveness. Here, the proposed LP is used to determine the weights of the proposed TransGuard-QNet. With this methodology, expected financial losses are reduced, and resource usage is optimized. Model performance is measured by several key metrics: accuracy (99.09%), precision (99.51%), recall (98.51%), F1-score (98.59%), and the economic impact. This approach has also outperformed existing methods with significant gains in detecting fraud as well as in operational efficiency and resource utilization. Thus, making it a promising candidate for deployment in dynamic transaction environments to manage financial risk.

Keywords Risk assessment · Machine learning · Fraud detection · Financial transaction · Linear programming optimization · Quantum computing

Extended author information available on the last page of the article

Published online: 15 November 2025

1 Introduction

The general incidence of financial fraud in transactions poses a question for both enterprises and individuals as the digital economy grows more developed. More so, the sophistication and incidence of fraud have been raised by technological advances and the elevated level of challenge exhibited by the attackers. One ill effect of fraudulent transactions, apart from money loss, is that it erodes consumers' confidence in financial institutions (Tayebi & El Kafhali, 2024). Therefore, much emphasis and resources are applied to the technology and processes that can identify and manage risks linked to those activities. Some of the conventional techniques for Fraud Detection (FD) mainly depend on rules based on which an event is claimed to be fraudulent if it crosses a certain threshold (Balcıoğlu, 2024). Despite their simplicity, these systems do not always learn new fraud trends and produce multiple unnecessary alarms. Due to the aforementioned limitations, new approaches such as Machine Learning (ML) (Vasudevan et al., 2024) have surged, and this is because it self-develops the model and looks for new patterns from historical data. Most supervised learning algorithms like Gradient Boosting Machines (GDBs) (Ding et al., 2024) and NNs (Wang et al., 2024a, 2024b) have been found to give very high accuracy in terms of detecting fraud. Likewise, unsupervised techniques such as AEs (Koo et al., 2024), Clustering (Huang et al., 2024), and independent component analysis (ICA) (Ali et al., 2024), etc., are applied to transaction profiles to identify abnormalities.

Supervised and unsupervised methods of ML are an active area of development in current times as the need to detect fraud has grown more complicated and requires a combination of both approaches (Patil et al., 2024). Collectively, these integrated frameworks have the objective of enhancing the precision of fraud risk prediction and reducing false alarms (Usman et al., 2024). Also, the latest feature engineering approaches are specifically designed. Other methods such as Genetic Algorithms (GA) (Nemati et al., 2024), and Bayesian approaches (Wang et al., 2024a, 2024b) for tuning parameters and thresholds of models are also rising in popularity. However, there are strong limitations in developing fraud detection systems even in today's society, some of which include the following reasons (Özbaltan, 2024). There is also a problem with the data skew, with legitimate transaction data being way higher compared to fraudulent transactions. This imbalance distorts generalization and makes it very hard for models to properly identify fraud cases (Ijiga et al., 2024). However, fraud schemes are ever evolving, and thus to keep up with these systems, data has to be updated, a computationally intensive process. A third issue is the need for timely detection since decisions in the financial industry might be needed within milliseconds. Last, but not least, the exact trade-off between precision and scalability still presents a strong barrier towards practical implementation (Chatterjee et al., 2024).

To overcome these challenges, the present work envisages the development of a novel integrated hybrid machine learning framework for predictive modelling, anomaly detection, and optimization. To achieve this, the framework combines supervised and unsupervised models with Linear Programming for resource allocation and decision optimization for improved FD and effective resource use. This approach also helps avoid losses since such frauds constantly change while enabling people to

guard money in the account from such fraud, making it an effective solution when the financial environment is always changing. The key contributions are:

- To develop a pioneering hybrid QML framework integrating supervised and unsupervised models with LP optimization.
- To design an advanced multi-model integration utilizing XGBoost, QNN, QAE, and clustering techniques.
- To extract multidimensional features spanning mathematical, behavioral, statistical, frequency, and correlation dimensions.
- To provide a dynamic risk scoring mechanism with adaptive threshold adjustment and intelligent resource allocation using LP optimization.

This article is structured as a recent literature on risk assessment and FD in Section 2. The implemented framework and the description are given in Section 3. Results and the significance are presented in Section 4. Section 5 ends the research.

2 Literature Study

2.1 Recent Research

Fatemi (2024) explored QML methods like QNNs, Quantum Support Vector Machines (QSVMs), and Quantum Principal Component Analysis (QPCA) for financial fraud detection, and also used a review-based method for assessing QML's role in improving accuracy and processing speed. The paper emphasized advantages like handling complex data proficiently. Drawbacks such as hardware immaturity and scalability issues.

Yu and Luo (2025) studied to develop a quantum-optimized deep learning model for effective financial fraud detection and also used a hybrid approach combining Deep Belief Networks (DBNs) with CNN, Long Short-Term Memory Networks (LSTM), and Graph Neural Networks (GNN) architectures, improved through quantum optimization. Advantages include improved precision, recall, faster training, and robust detection of complex fraud patterns. Moreover, challenges include high implementation complexity and the need for advanced quantum computing resources.

Nur (2025) examined to the development of a quantum-deep learning hybrid model for real-time fintech fraud detection and also used QSVMs for quantum-enhanced feature mapping and LSTM networks to capture temporal transaction patterns. The method displayed better detection accuracy, speed, and generalization over classical models. However, limitations include dependency on quantum hardware availability and integration complexity.

Bello et al. (2023) suggested Supervised and Unsupervised Learning, DL, Natural Language Processing (NLP), and hybrid models for real-time fraud detection in U.S. financial transactions. The aim is to progress fraud inhibition and operational efficiency through AI-driven systems also addressing ethical, regulatory, and scalability issues with real-life case studies. Moreover, the model faced restrictions related to data privacy and limited cross-border fraud detection.

Akash et al. (2024) examined fraud prevention using Artificial Neural Network (ANN), statistical techniques, and rule-based approaches, also aimed to combine multiple detection strategies to reduce False Positives and enhance security. The advantage is the evaluation of current systems and best-practice recommendations. A major limitation was the dependence on a sample dataset, restricting broader applicability.

Malik et al. (2024) suggested ML models like Logistic Regression (LR), Decision Tree (DT), CNNs, RNNs, and NLP to detect fraud, also aimed to classify both known and emerging fraud patterns in financial transactions. Advantages included real-time detection and the use of both structured and unstructured data. Limitations involved data privacy, model interpretability, and training quality.

Udayakumar et al. (2023) proposed a hybrid model combining SVM for feature extraction and Feedforward Neural Network (FFNN) for learning transaction patterns, and also leads an increase of accuracy and precision in fraud detection using ensemble techniques. The approach improved detection performance on labelled datasets. However, reliance on a single dataset limited its generalization across varied financial contexts.

Goriparthi (2023) evaluated AI-augmented data mining using DT, SVMs, and DL for large-scale fraud detection and also aimed to uncover hidden fraud patterns in vast financial datasets. Advantages such as improved detection speed and accuracy compared to traditional methods. The limitation was the focus on specific models, affecting generalizability across broader domains.

2.2 Problem Statement

Table 1 demonstrates the aim and limitations of recent literature on FD in financial transactions using various ML methods. Financial transaction risk assessment based on ML in financial transactions is capable of presenting advanced probabilities in fraud detection, and this is through real-time anomaly detection, pattern recognition, and charges of risk charges. The approach allows for the evaluation of the data in multiple directions, greatly minimizing the number of false positives, efficient use of resources, and detailed classification of risks. These systems use modern approaches such as QNNs, ensemble learning, and probabilistic modeling and analyze enormous transactional data sets with uncompromised computational rationality and predictive precision. The current state of technological development presents several core issues with ML-based FD systems, including the presence of bias in training datasets, computational difficulties, and the dynamic nature of fraud. The models rely strictly on labeled training data and are sensitive to the quality of data, as well as have a problem with new fraud patterns that have not been observed in historical data. Besides, there are problems of computationally expensive procedures or high computational costs, issues of privacy for the financial data, as well as the problem of retraining the model to maintain the effectiveness of the detection rate in constantly evolving conditions within the financial industry. Thus, there exists room for further improvements in FD using innovative approaches with the objective of accurate predictions. Aim and Limitations of Recent Literature on FD in Financial Transactions Using Various ML Methods.

Table 1 Aim and limitations of recent literature on FD in financial transactions using various ML methods

Authors/Year	Methods	Aim	Advantages	Limitations
Fatemi (2024)	QNNs, QSVMs, QPCA	To evaluate QML's role in fraud detection	Handled complex data	Scalability issues
Yu and Luo (2025)	DBNs with CNN, LSTM, GNN with Quantum Optimization	To develop a quantum-optimized deep learning model	Achieved high precision, robust for complex fraud patterns	High complexity
Nur (2025)	QSVMs, LSTM	To improve Real-time fintech fraud detection with hybrid quantum-deep learning	Reached high accuracy	Integration complexity
Bello et al. (2023)	Supervised/Unsupervised ML, DL, NLP, Hybrid Models	To improve real-time fraud detection and operational efficiency	Scalability issues addressed	Data privacy concerns
Akash et al. (2024)	ANN, Statistical Methods, Rule-based Systems	To integrate strategies to reduce false positives and improve security	Evaluation of current systems, practical recommendations	Sample dataset limits broad application
Malik et al. (2024)	LR, DT, CNN, RNN, NLP	To classify known and emerging fraud patterns	Real-time detection uses both structured and unstructured data	Data privacy, model interpretability, and training data quality
Udayakumar et al. (2023)	SVM with FFNN Ensemble	To enhance detection accuracy using a hybrid model	Improved precision, pattern learning	A single dataset restricts generalization
Goriparthi (2023)	DT, SVM, DL	To detect fraud in large-scale financial datasets	Obtained high accuracy	Narrow focus on specific models

3 A Novel Risk Assessment and Fraud Detection Model

3.1 Proposed Architecture

For financial transaction risk assessment and prediction, this study presents TransGuard-QNet, an integrated hybrid QML framework that blends supervised and unsupervised models with LP for decision optimization. Initially, the raw input data is preprocessed with mean imputation (MI) to handle missing values, Min–Max scaling for standardization, Low-Pass Filtering (LPF) for noise removal, and the log transformation to ensure non-negative outputs. Additionally, for increased robustness, the suggested model uses iForest to monitor outliers and Z-score thresholding to identify values that are beyond the threshold. Moreover, mathematics, behaviours, statistics, frequency, and correlation features are the main components of feature engineering. QAEs and k-means clustering algorithms are used to identify anomalies, while iXGBoost is employed to deal with imbalanced data and learn nonlinear interactions. Its tree-based, interpretable structure helps interpret decision boundaries in fraud classification. QNNs are used because of their better ability to detect quantum correlations and learn intricate data structures, which are beneficial for detecting subtle patterns of fraud in large sets of transactions. QAE is employed in unsupervised

anomaly detection, can recreate normal patterns of transactions and detect anomalies with high sensitivity, and k-Means Clustering incorporated as a lightweight unsupervised module to cluster similar transactions and identify outliers that do not belong to established clusters. Here, the weights of the proposed TransGuard-QNet are optimized via implemented LP optimization. This model optimizes resource use and minimizes anticipated financial losses. Figure 1 depicts the overview of the proposed model.

3.2 Preprocessing

The primary goal of implementing the preprocessing phase is to clean and prepare the raw transactional data. Besides, this phase reduces noise, handles missing values, and transforms the data into a standardized format, which ensures it is in a suitable format for subsequent ML and optimization processes.

3.2.1 MI

It replaces missing values in the dataset with its mean value to ensure completeness and avoid errors during model training (Bhushan et al., 2024). For a feature x with n non-missing values, the MI is estimated as per Eq. (1), where x_j represents the non-missing values of feature x.

$$x_i = \sum_{j=1}^{n} x_j \tag{1}$$

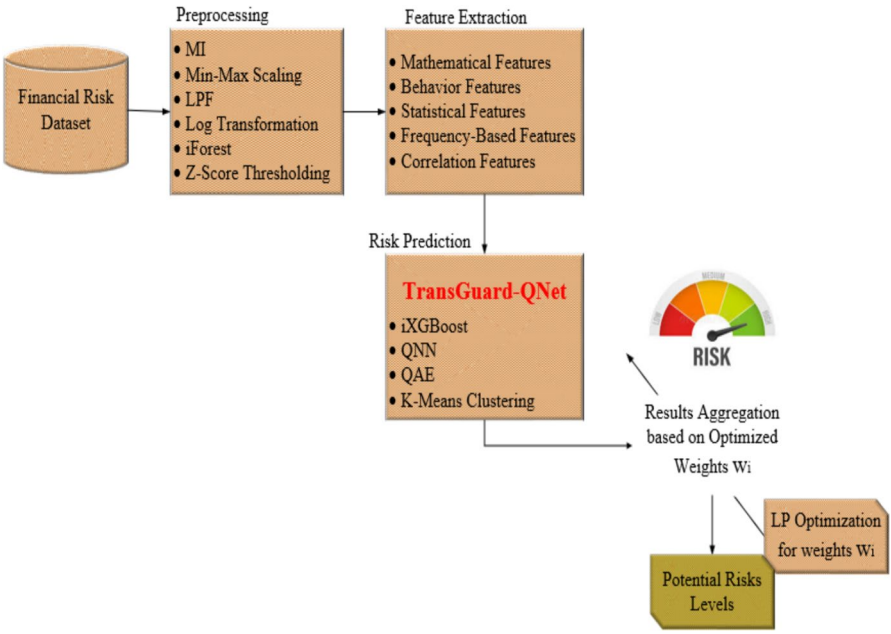


Fig. 1 Overview of proposed model

The imputed value x_i is then used to replace all missing entries for the feature.

3.2.2 Min–Max Scaling

It transforms features to a uniform range, typically [0, 1], to prevent features with large ranges from dominating those with smaller ranges (Rahmad Ramadhan & Anne Mudya, 2024). Equation (2) shows the min–max scaling to standardize the data, where x indicates the original feature value, x_{min} and x_{max} refer to the minimum and maximum values of the feature, and $x\prime$ states the scaled value.

$$x' = \frac{\mathrm{x} - x_{min}}{(x_{max} - x_{min})} \tag{2}$$

3.2.3 LPF Noise Removal

It removes high-frequency noise from the dataset, particularly in time-series data, by allowing only low-frequency components (He et al., 2024). Normally, an LPF smooths the data by attenuating high-frequency variations. The smoothed value S_t for time t using a simple moving average filter is calculated using Eq. (3), in which k is the window size, and x_i addresses the original data points.

$$S_t = \frac{1}{k} \sum\nolimits_{i=t-k+1}^{t} x_i \tag{3}$$

3.2.4 Log Transformation

It is employed to handle skewed data by compressing large values and ensuring all outputs are non-negative (Guerand & Imbert, 2023). Equation (4) explains the applied log transformation, in which x stands for the original feature value, $\log$ refers to the natural logarithm, and x' signifies the transformed value.

$$x' = \log(1 + x) \tag{4}$$

3.2.5 iForest Outlier Monitoring

It is applied to detect and isolate outliers by measuring their degree of isolation in a dataset. Typically, iForest builds a binary tree to isolate data points. Points that are easily isolated are considered outliers (Yousef et al., 2024). Equation (5) exhibits the outlier score estimation, in which $E(h(x))$ specifies average path length for the point x, $c(n)$ denotes average path length in a binary tree for n observations, and $s(x, n)$ represents anomaly score (closer to 1 indicates an outlier).

$$s(x, n) = 2^{\left(-\frac{E(h(x))}{c(n)}\right)} \tag{5}$$

3.2.6 Z-Score Thresholding for Outlier Removal

Points with scores above a specified threshold are flagged as outliers (Yaro et al., 2024). For this estimation, Z-score thresholding is used. It identifies extreme values that deviate significantly from the mean as shown in Eq. (6), where x signifies a data point, μ refers to mean, and σ specifies standard deviation (SD).

$$z = \frac{x - \mu}{\sigma} \tag{6}$$

Values exceeding a predefined threshold $|z| > k$ are classified as outliers and are removed. Now, the dataset is prepared for accurate and reliable ML model training and optimization.

3.3 Feature Engineering

Feature engineering aims to transform raw data into features that enhance the predictive power of ML models. For this procedure, several significant features are extracted. It includes:

3.3.1 Mathematical Features

It is derived using algebraic and analytical operations (e.g., sum, product, and ratios) on transaction data to capture intrinsic relationships, like average transaction amount per user.

Transaction Amount Logarithmic Transformation Transaction amounts often exhibit skewness, where a small percentage of transactions have significantly high values. Logarithmic transformation (Zhu et al., 2024) reduces this skewness by stabilizing variance and ensuring the data is less sensitive to outliers as shown in Eq. (7), in which adding 1 avoids undefined values for $log\,(0)$.

$$LogTransformed_{Feature} = \log\,(Amount + 1) \tag{7}$$

It reduces the impact of large transaction values.

Time Series-Based Moving Average The moving average (MA) (Deepan et al., 2024) smooths transaction data over time by capturing underlying trends or seasonal patterns. This feature is crucial for detecting deviations from expected behavior. Equation (8) explains the MA procedure, where n addresses time window size, and x_{t-i} represents the transaction at time $t - i$.

$$MA_t = \frac{1}{n} \sum\nolimits_{i=1}^{n} x_{t-i} \tag{8}$$

It helps identify anomalies, such as spikes in transaction amounts, and provides a smoothed representation of trends.

3.3.2 Behavioral Features

It captures patterns of user activity, such as spending habits, time of transactions, or preferred categories, by highlighting deviations from normal behavior.

Frequency of Transactions It measures the number of transactions made by an entity within a specific timeframe. Significant increases or decreases in frequency indicate anomalous behavior as indicated in Eq. (9).

$$Frequency \frac{= Total\ Transactions}{Time\ Interval} \tag{9}$$

Here, high frequency indicates automated or fraudulent transactions, and low frequency indicates inactivity or infrequent usage.

Historical Anomalies It identifies patterns in past transactions that deviate significantly from typical behavior. Transactions are assessed for periodic irregularities by examining seasonal patterns (e.g., weekly or monthly) using Fourier Transform (FT) as given in Eq. (10), in which $x\,(t)$ represents the transaction amount over time, and $X\,(f)$ reveals dominant periodic trends.

$$X\,(f) = x\,(t)\,e^{-2\pi i f t} \tag{10}$$

Here, deviations from expected frequencies indicate anomalies.

3.3.3 Statistical Features

It includes metrics like mean, median, variance, and skewness to quantify the distribution and variability of transactional data.

Central Tendency Mean transaction amount (μ) calculates the average transaction amount serves as a baseline for typical transaction size as specified in Eq. (11). It indicates the typical transaction amount for an account.

$$\mu = \frac{1}{n} \sum\nolimits_{i=1}^{n} x_i \tag{11}$$

Median transaction amount finds the median M value of the dataset with an odd number of transactions (N), the median is the value at the middle position in the sorted list as signified in Eq. (12).

$$\mathrm{M} = x\left(\frac{N+1}{2}\right) \tag{12}$$

When the dataset has an even number of transactions (N), the median is calculated as the average of the two central values shown in Eq. (13).

$$\mathrm{M} = \frac{1}{2}\left(x\left(\frac{N}{2}\right) + x\left(\frac{N}{2}+1\right)\right) \tag{13}$$

3.3.4 Dispersion Features

SD (σ) measures variability in transaction amounts as expressed in Eq. (14), in which high variability indicates inconsistent behavior.

$$\sigma = \sqrt{\frac{1}{n}\sum\nolimits_{i=1}^{n}\left(x_i - \mu\right)^2} \tag{14}$$

Range expresses the difference between the maximum and minimum transaction amounts as formulated in Eq. (15).

$$Range = \max\left(x_i\right) - \min\left(x_i\right) \tag{15}$$

Interquartile Range (IQR) uses the spread of the middle 50% of data, robust to outliers as specified in Eq. (16), where Q_3 refers to $the 75^{th}$ percentile and Q_1 addresses the 25th percentile.

$$IQR = Q_3 - Q_1 \tag{16}$$

3.3.5 Distributional Features

Skewness measures asymmetry in the distribution of transaction amounts as indicated in Eq. (17).

$$Skewness = \frac{\frac{1}{n}\sum_{i=1}^{n}\left(x_i - \mu\right)^3}{\sigma^3} \tag{17}$$

Kurtosis measures the "tailedness" of the distribution by indicating the presence of extreme values as illustrated in Eq. (18).

$$\mathrm{Kurtosis} = \frac{\frac{1}{n}\sum_{i=1}^{n}\left(x_i - \mu\right)^4}{\sigma^4} - 3 \tag{18}$$

3.3.6 Temporal Features

Interarrival time measures the time gaps between consecutive transactions t as represented in Eq. (19).

$$Interarrival\ Time = t_{i+1} - t_i \tag{19}$$

Peak transaction hours identify hours with the highest transaction frequency.

3.3.7 Frequency-Based Features

It tracks the count or rate of specific events (e.g., transactions per day or fraudulent attempts) through recurring or unusual patterns.

Transaction Count It counts the total number of transactions within a period. Notably, high transaction counts in short timeframes indicate automation.

Proportion of High-Risk Transactions It indicates the fraction of transactions flagged as high-risk historically, as given in Eq. (20).

$$Proportion \frac{= High\ Risk\ Transactions}{Total\ Transactions} \tag{20}$$

3.3.8 Correlation Features

It analyzes the linear relationship between transaction patterns and external factors (e.g., location, device type, or time). These correlations help identify deviations from typical transaction behavior, potentially flagging anomalies. The correlation coefficient r, between two variables X (transaction patterns) and Y (external factors) is given in Eq. (21), where X_i refers to individual transaction values in the dataset, Y_i addresses associated value of the external factor (location or time), X' stands for mean of transaction values X, Y' indicates mean of external factor values Y, and N denotes data point count.

$$r = \frac{\sum_{i=1}^{n} (X_i - X_i')(Y_i - Y_i')}{\sqrt{\sum_{i=1}^{n} (X_i - X_i')^2} \sqrt{\sum_{i=1}^{n} (Y_i - Y_i')^2}} \tag{21}$$

After the feature engineering phase, the prepared data undergoes advanced ML and quantum-based processes for risk prediction and anomaly detection.

3.4 Risk Prediction and Fraud Detection Via Proposed TransGuard-QNet

The proposed TransGuard-QNet combines XGBoost and QNNs to predict risk probabilities and employs QAEs and k-means clustering to detect anomalies. These pre-

dictions are aggregated into a final result and optimized using weights to classify transactions as low, medium, or high risk for effective decision-making.

3.4.1 Risk Prediction

XGBoost Normally, XGBoost (Luo et al., 2024) is a supervised ML algorithm used to predict the risk probability of transactions. It trains on the engineered features X (mathematics, behavior, statistics) and the target variable Y (fraudulent or non-fraudulent). The risk probability R_p for a transaction is computed using the learned model described in Eq. (22), in which Θ_{XGB} represents the learned parameters of the XGBoost model.

$$R_p = f_{XGB}\left(X, \Theta_{XGB}\right) \tag{22}$$

iXGBoost introduces enhancements to the standard Gradient Boosting framework by focusing on computational efficiency, regularization, and improved handling of sparse data. The model optimizes a custom objective function by adding regularization terms to prevent overfitting and improve generalization. The objective function in XGBoost is expressed in Eq. (23), where $\ell\left(yi, y_i'\right)$ addresses the log loss function measuring the difference between the true label y_i and the predicted value y_i'.

$$L\left(\Theta\right) = \sum\nolimits_{i=1}^{n} \ell\left(yi, y_i'\right) + \sum\nolimits_{t=1}^{T} \Omega\left(f_t\right) \tag{23}$$

To improve the detection of financial fraud in severely imbalanced transaction datasets, the proposed iXGBoost framework combines a regularized objective function with a second-order optimization approach. Here, $\Omega\left(f_t\right)$ signifies the regularization term which is defined in Eq. (24), where T denotes leaf count in the tree, w_j indicates leaf weights, γ stands for the regularization parameter controlling model complexity, and λ states the L2 regularization term.

$$\Omega\left(f_t\right) = \gamma T + \frac{1}{2}\lambda \sum\nolimits_{j=1}^{T} w_j^2 \tag{24}$$

Ensemble-based prediction model improved by summing contributions from all trees as shown in Eq. (25), in which $f_t\left(x_i\right)$ represents the output of the tth tree for input x_i, and the final prediction $\widehat{y}_i$ denotes the transaction x_i as the sum of outputs from all trees. This ensemble method combines many weak learners, which is particularly helpful in risk-scoring situations, where fraud activity is frequently non-linear and spread throughout many features.

$$y_i' = \sum\nolimits_{t=1}^{T} f_t\left(x_i\right) \tag{25}$$

iXGBoost uses a second-order Taylor expansion of the loss function to efficiently minimize the loss during training. This balances predictive performance with regu-

larization by incorporating both the gradient and curvature (Hessian) of the loss, as stated in Eq. (26), where g_i and h_i address the first and second derivatives of the loss function, which are estimated using Eq. (27) and (28) in order. This steers the tree construction by determining how far from the mark each prediction is and how sure the model is of the error. This is especially important in fraud detection, where false negatives misclassifying fraudulent cases have a high monetary cost.

$$L^t = \sum\nolimits_{i=1}^{n} \left[g_i f_t\left(x_i\right) + \frac{1}{2} h_i f_t(x_i)^2 \right] + \Omega\left(f_t\right) \tag{26}$$

$$g_i = \frac{\partial \ell\left(yi, y_i'\right)}{\partial y_i'} \tag{27}$$

$$h_i = \frac{\partial^2 \ell\left(yi, y_i'\right)}{\partial y_i'^2} \tag{28}$$

Learning rate for incremental refinement is defined as per Eq. (29), where Learning rate (η) controls the contribution of each tree by allowing better fine-tuning. This prevents overfitting, particularly in unstable financial conditions, by continuously narrowing the fraud prediction boundary.

$$y_i' = \sum_t = 1^T \eta f_t\left(x_i\right) \tag{29}$$

3.4.2 QNN

QNNs (Schatzki et al., 2024) process the same input features to compute an alternative risk probability P_q by using quantum states for efficient learning. QNNs utilize principles of quantum mechanics to process information efficiently by representing data as quantum states. This capability enables them to learn complex patterns and relationships in datasets with potentially fewer resources compared to classical neural networks. The risk probability P_q computed by QNNs is expressed in Eq. (30), where X is the input feature vector, and Θ_{QNN} addresses a set of parameters in the QNN, such as weights and biases, that govern the transformation of quantum states.

$$P_q = f_{QNN}\left(X, \Theta_{QNN}\right) \tag{30}$$

It is computed through the following process. The engineered input features X are encoded into quantum states $|\psi\left(X\right)|$ using quantum feature maps as indicated in Eq. (31), in which $U_{enc}\left(X\right)$ defines a Quantum unitary transformation encoding the input into a quantum Hilbert space.

$$|\psi\left(X\right)\rangle = U_{enc}\left(X\right)|0\rangle \tag{31}$$

A series of quantum gates, parameterized by Θ_{QNN}, are applied to the encoded state to model complex patterns as illustrated in Eq. (32).

$$|\phi(X)\rangle = U_{QNN}(\Theta_{QNN})|\psi(X)\rangle \tag{32}$$

The quantum state $|\phi(X)\rangle$ is measured to extract probabilities as given in Eq. (33), in which $|y\rangle$ represents the target state for risk prediction.

$$P_q = |\langle y|\phi(X)\rangle|^2 \tag{33}$$

To optimize Θ_{QNN}, a cross-entropy loss function is minimized as shown in Eq. (34), in which y_i refers to the true label, and n stands for the number of samples.

$$L(\Theta_{QNN}) = -\sum_{i=1}^{n} y_i \log(P_q) + (1 - y_i)\log(1 - P_q) \tag{34}$$

3.4.3 QAE

QAEs (Du & Tao, 2024) are used to compress the input features and reconstruct them to identify anomalies. If the reconstruction error E_r exceeds a threshold δ, the transaction is flagged as anomalous, as illustrated in Eq. (35), where X refers to the original input feature vector, $f_{QAE}(X)$ states reconstructed feature vector from the QAE, and $\|\cdot\|$ denotes norm used to measure the difference (L_2-norm).

$$E_r = \|X - f_{QAE}(X)\| \tag{35}$$

A transaction is flagged as anomalous if the reconstruction error exceeds a predefined threshold δ as given in Eq. (36).

$$E_r > \delta \tag{36}$$

The proposed QAE is computed by the following procedure. Engineered input feature X is mapped to quantum states $|\psi(X)\rangle$ using quantum feature maps as expressed in Eq. (37).

$$\psi(X)\rangle = U_{enc}(X)|0\rangle \tag{37}$$

QAEs reduce the dimensionality of the input features using parameterized quantum gates, projecting the quantum state into a compressed latent space as formulated in Eq. (38).

$$|\phi latent(X)\rangle = U_{compress}(\Theta_{QAE})|\psi(X)\rangle \tag{38}$$

The compressed state is expanded back into the original space as indicated in Eq. (39).

$$|\phi reconstructed\,(X)\rangle = U_{reconstruct}\,(\Theta_{QAE})\,|\phi_{latent}\,(X)\rangle \tag{39}$$

The reconstructed quantum state is measured and converted to a classical output $f_{QAE}\,(X)$. Finally, the difference between X and $f_{QAE}\,(X)$ is computed as E_r. If $E_r > \delta$, the transaction is flagged as anomalous. Finally, the anomaly score derived from QAEs, based on reconstruction error, is decided using Eq. (40).

$$S_{QAE} = \begin{cases} 1 & E_r > \delta \\ 0 & o.w \end{cases} \tag{40}$$

3.4.4 K-Means Clustering

The feature space is clustered into k groups based on similarity using k-Means clustering (Zhang et al., 2024). Each cluster is represented by its center C_k, calculated as the mean of all feature vectors X_i belonging to that cluster, as referred in Eq. (41), where n_k stands for the number of points in $thek^{th}$ cluster, and X_i addresses the feature vector of the ith transaction.

$$C_k = \frac{1}{n_k} \sum\nolimits_{X_i \in Cluster_k} X_i \tag{41}$$

Transactions that fall outside the main clusters or within low-density clusters are flagged as anomalous, as explained in Eq. (42), where D_i refers to the distance of the ith transaction to the nearest cluster center C_k, and $\|\cdot\|$ denotes the Euclidean norm used to measure similarity.

$$D_i = \min_k \|X_i - C_k\| \tag{42}$$

A transaction is flagged as anomalous if its distance D_i exceeds a predefined threshold τ, as shown in Eq. (43), where τ defines the threshold determined during training to distinguish normal points from anomalies.

$$D_i > \tau \tag{43}$$

Finally, the anomaly score derived from k-means clustering, based on distance to the nearest cluster center, is estimated based on Eq. (44).

$$S_{kMeans} = \begin{cases} 1 & D_i > \tau \\ 0 & o.w \end{cases} \tag{44}$$

3.4.5 Results Aggregation

The outputs are combined using optimized weights $w_1, w_2, w_3, and w_4$ determined through training and cross-validation to maximize accuracy performance. The final

output S_{final} is computed in Eq. (45), where $w_1 + w_2 + w_3 + w_4 = 1$, ensuring the weights sum to 1.

$$S_{final} = w_1 R_p + w_2 P_q + w_3 S_{QAE} + w_4 S_{kMeans} \tag{45}$$

Based on the weights w_i transactions are classified into low, medium, or high risk. Here, the objective is to optimize the weights so that the composite score $S_{composite}$ aligns with the labeled risk levels in the dataset as indicated in Eq. (46).

$$S_{composite} = \sum\nolimits_i w_i \cdot S_i \tag{46}$$

3.5 Proposed LP for Weights Alignment

Here, the weights for the proposed TransGuard-QNet are determined using the proposed LP. Normally, LP (Ghaithan et al., 2024) is a mathematical optimization technique used to maximize or minimize a linear objective function, subject to linear equality and inequality constraints. The primary goal of LP in this context is to optimize the weights w_1, w_2, w_3, w_4 such that the composite score $S_{composite}$ aligns with the labeled risk levels (low, medium, high). The objective is to minimize the error between $S_{composite}$ and the target risk levels, subject to weight constraints shown in Eq. (47). The risk alignment constraints are expressed in Eq. (48), and error constraints are given in Eq. (49), where T_1 and T_2 are risk thresholds.

$$\sum w_i = 1 \text{ and } w_i \geq 0 \forall i = 1,2,3,4 \tag{47}$$

$$Risk\ Level = \begin{cases} \text{Low} & S_{composite} < T_1 \\ \text{Medium} & T_1 \leq S_{composite} < T_2 \\ High & S_{composite} \geq T_2 \end{cases} \tag{48}$$

$$e^j \geq S^j_{composite} - R^j; e^j \geq R^j - S^j_{composite} \tag{49}$$

The objective is expressed in Eq. (48),

$$Minimize : \sum\nolimits_j e^j \tag{50}$$

Now, the LP formulation is appropriately structured for optimizing the weights to align the composite score with the labeled risk levels. Algorithm 1 shows the pseudocode of the proposed TransGuard-QNet. Table 2 summarizes the parameter settings of the proposed TransGuard-QNet.

Algorithm 1 Pseudocode of proposed TransGuard-QNet

Step 1	**Initialization** Initialize XGBoost, QNN, QAE, and K-Means models
Step 2	**Load Dataset** Load dataset with engineered features X and target variable Y
Step 3	**Train all the ML models** For each transaction X in the dataset
Step 4	Compute Risk Probability from XGBoost as per Eq. (22)
Step 5	Compute Risk Probability from QNN as per Eq. (30)

Step 6	Compute Reconstruction Error from QAE using Eq. (35)
	If $E_r > \delta$
	$S_{QAE} = 1$
	Else
	$S_{QAE} = 0$
	End if

Step 6	Compute Anomaly Score from K-Means via Eq. (42)
	If $D_i > \tau$
	$S_{KMeans} = 1$
	Else
	$S_{KMeans} = 0$
	End if

Step 7	Aggregate results using weighted sum using Eq. (45)

Step 8	Classify Transactions based on the final score
	if $S_{composite} < T_1$
	Risk Level = "Low"
	Else if $T_1 \leq S_{composite} < T_2$
	Risk Level = "Medium"
	Else if $S_{composite} \geq T_2$
	Risk Level = "High"
	End if
	End for

Step 9	Optimization of Weights (LP formulation) as per Eq. (48)
Step 10	Return Risk Level for each transaction

Table 2 Parameter settings of proposed TransGuard-QNet

Component	Parameter	Value
Data	Input dataset with transactions	Engineered Features
Threshold for reconstruction error from QAE	δ	0.05
Threshold for distance in K-Means clustering anomaly detection	τ	0.1
Aggregation Weights	w_1, w_2, w_3, w_4	Weights for the final aggregated risk score
LP solver Implementation	LP_solver	Linear Programming solver for weight optimization
Weight for XGBoost risk prediction	weights [0]	0.25
Weight for QNN risk prediction	weights [1]	0.25
Weight for QAE anomaly detection	weights [2]	0.25
Weight for k-Means anomaly detection	weights [3]	0.25
Low-risk threshold	T_1	0.3
Medium risk threshold	T_2	0.7

4 Simulation Results

4.1 Simulation Setup

The proposed risk prediction model using the implemented TransGuard-QNet was developed via Python on Intel core® i5 processor @2.6 GHz, 16 GB RAM, 64-bit OS. For experimentation, the Financial_Risk dataset is used, which is accessible via https://www.kaggle.com/datasets/preethamgouda/financial-risk [Access date: 11–12–2024]. Here, the Financial Risk Assessment Dataset provides key financial indicators that help assess an individual's financial stability and risk. It includes data on income, credit score, loan amount, employment status, and other factors, enabling financial institutions to evaluate creditworthiness and predict potential financial risks. By incorporating variables like loan purpose and risk rating, the dataset helps model financial behaviors and guide decisions related to lending, insurance, and financial planning. The dataset's real-world imbalances and missing values further simulate practical financial scenarios, offering insights into diverse financial profiles. The efficacy of the proposed TransGuard-QNet is evaluated through a comparative analysis with baseline models such as iXGBoost, QNN, QAE, and K-means clustering. Besides, benchmark ML models such as DT, RF, KNN, and LR (Dong et al., 2024) are compared with the proposed model.

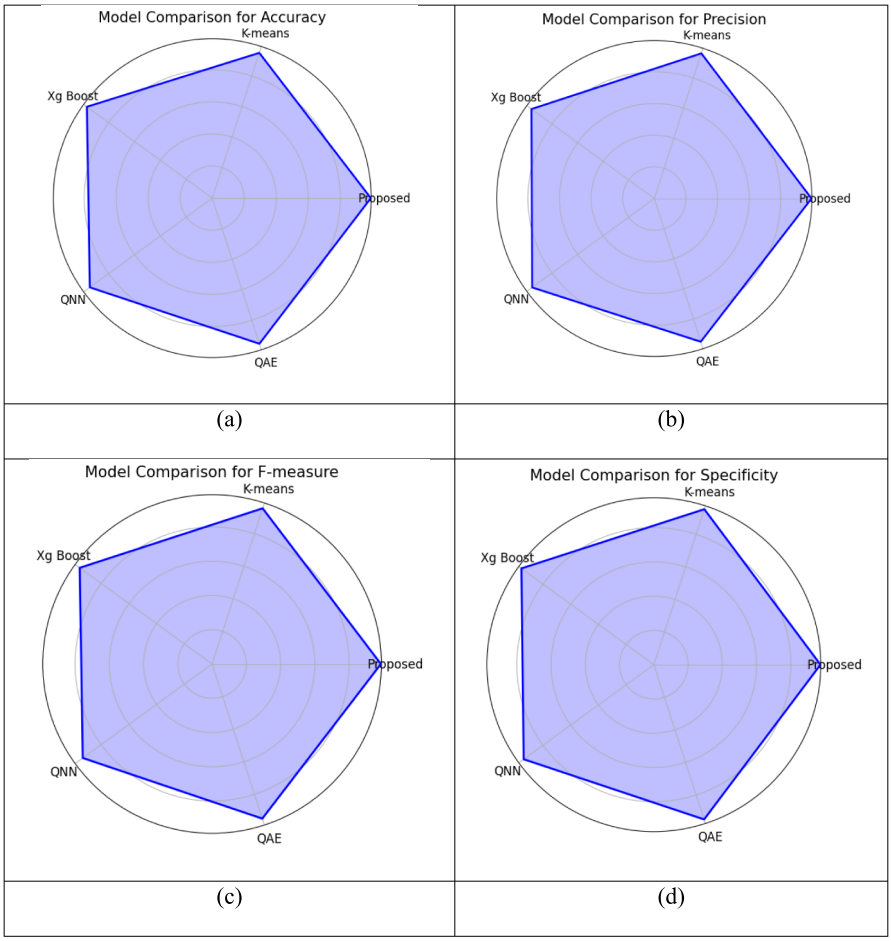


Fig. 2 Performance of Proposed TransGuard-QNet over Baseline Models for (**a**) Accuracy, (**b**) Precision, (**c**) F1-Score, (**d**) Specificity, (**e**) Sensitivity, (**f**) NPV, (**g**) MCC, (**h**) FPR, and (**i**) FNR

4.1.1 Algorithmic Analysis

The attained results and the advancements of the proposed TransGuard-QNet are presented here. Figure 2 shows the performance of the proposed TransGuard-QNet over baseline models. Here, the data is split into 80:20 learning samples. From Fig. 2, it is clear that the proposed TransGuard-QNet highlights its superior accuracy and anomaly detection capabilities. The results confirm TransGuard-QNet's effectiveness in achieving higher detection precision and lower false positives. Similarly, Fig. 3 exhibits the performance of the proposed TransGuard-QNet over SOTA models. The results demonstrate higher detection accuracy and better adaptability to complex scenarios. This validates the developed TransGuard-QNet's robustness and effectiveness over existing SOTA models.

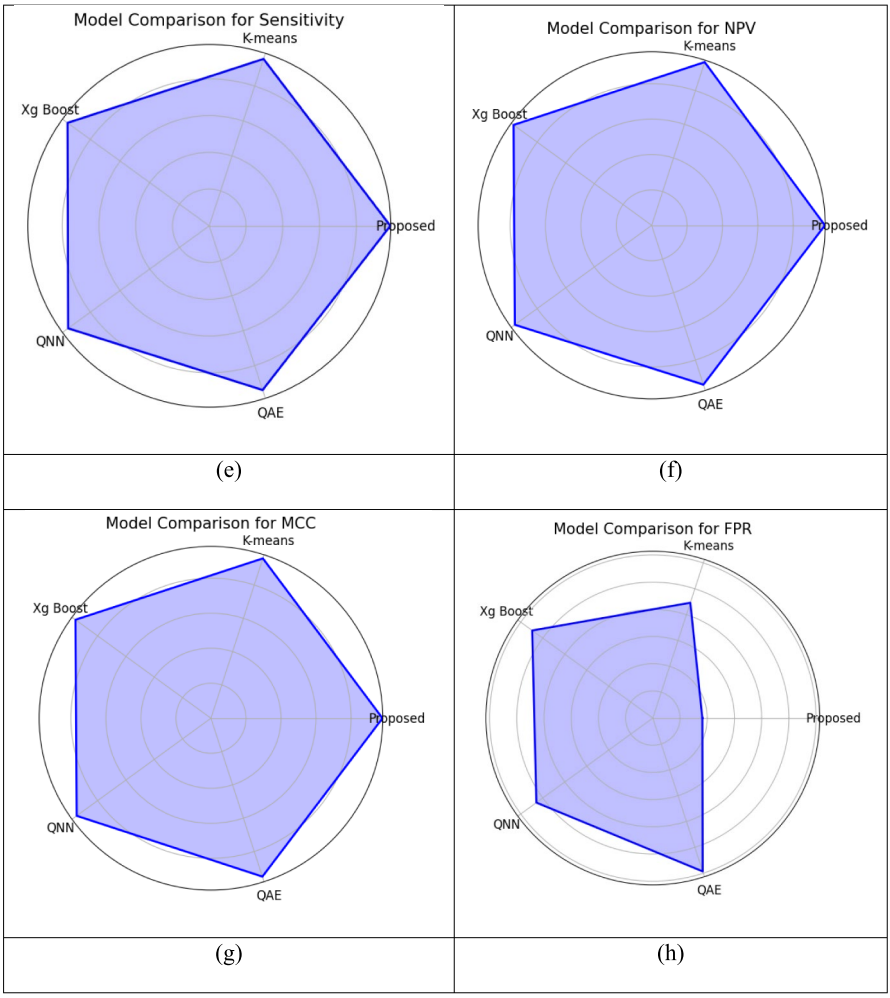
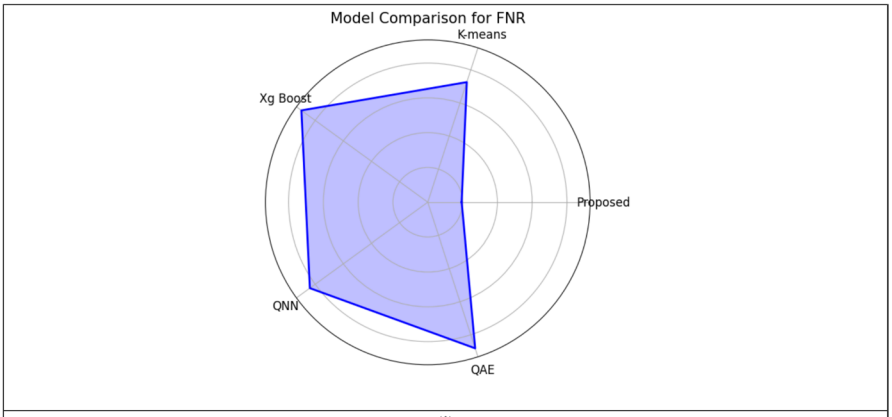


Fig. 2 (continued)

In Table 3, the proposed TransGuard-QNet demonstrates superior performance compared to baseline models across key evaluation metrics. It achieves the highest accuracy (99.095%), precision (99.512%), and F1-score (98.591%), alongside excellent specificity (97.291%) and sensitivity (98.519%), indicating robust and reliable detection. Notably, it also excels in metrics such as Negative Predictive Value (NPV) (97.893%) and Mathew's Correlation coefficient (MCC) (97.973%) while maintaining the lowest False Positive Rate (FPR, 0.0182) and False Negative Rate (FNR, 0.00974). In contrast, baseline models like K-means, XGBoost, QNN, and QAE lag in all metrics, showcasing the proposed model's effectiveness and efficiency in handling complex tasks.



(i)

Fig. 2 (continued)

On the other hand, Table 4 explains the performance of the proposed TransGuard-QNet over benchmark models. The proposed TransGuard-QNet outperforms the traditional models, such as RF, DT, KNN, and LR, in terms of various metrics. With an accuracy of 99.095% and precision of 99.512%, it demonstrates superior classification capability. Additionally, it achieves the highest F-measure (98.591%), specificity (97.291%), sensitivity (98.519%), NPV (97.893%), and MCC (97.973%), ensuring robust performance across various metrics. Moreover, its FPR (0.0182) and FNR (0.00974) are significantly lower than those of the baseline models, showcasing its reliability and precision in handling diverse data characteristics.

Figure 4 displays the training and validation accuracy and loss of the proposed model over 100 epochs. The training accuracy steadily increases, approaching 99.09%, while validation accuracy stabilizes around 97%, indicating excellent generalization. In the loss graph, both training and validation losses decrease rapidly in the initial epochs, with training loss approaching zero, showcasing effective learning. The relatively stable validation loss after early epochs suggests minimal overfitting and robust performance on unseen data.

4.2 Scalability and Efficiency

The proposed iXGBoost framework exhibits excellent scalability and computational efficiency, which makes it suitable for deployment in real-time and large-scale financial settings. The second-order Taylor approximation enables faster convergence during training by utilizing both gradient and Hessian information. The model also includes tree pruning and regularization methods that minimize unnecessary complexity, allowing for rapid inference times even with high amounts of transaction data.

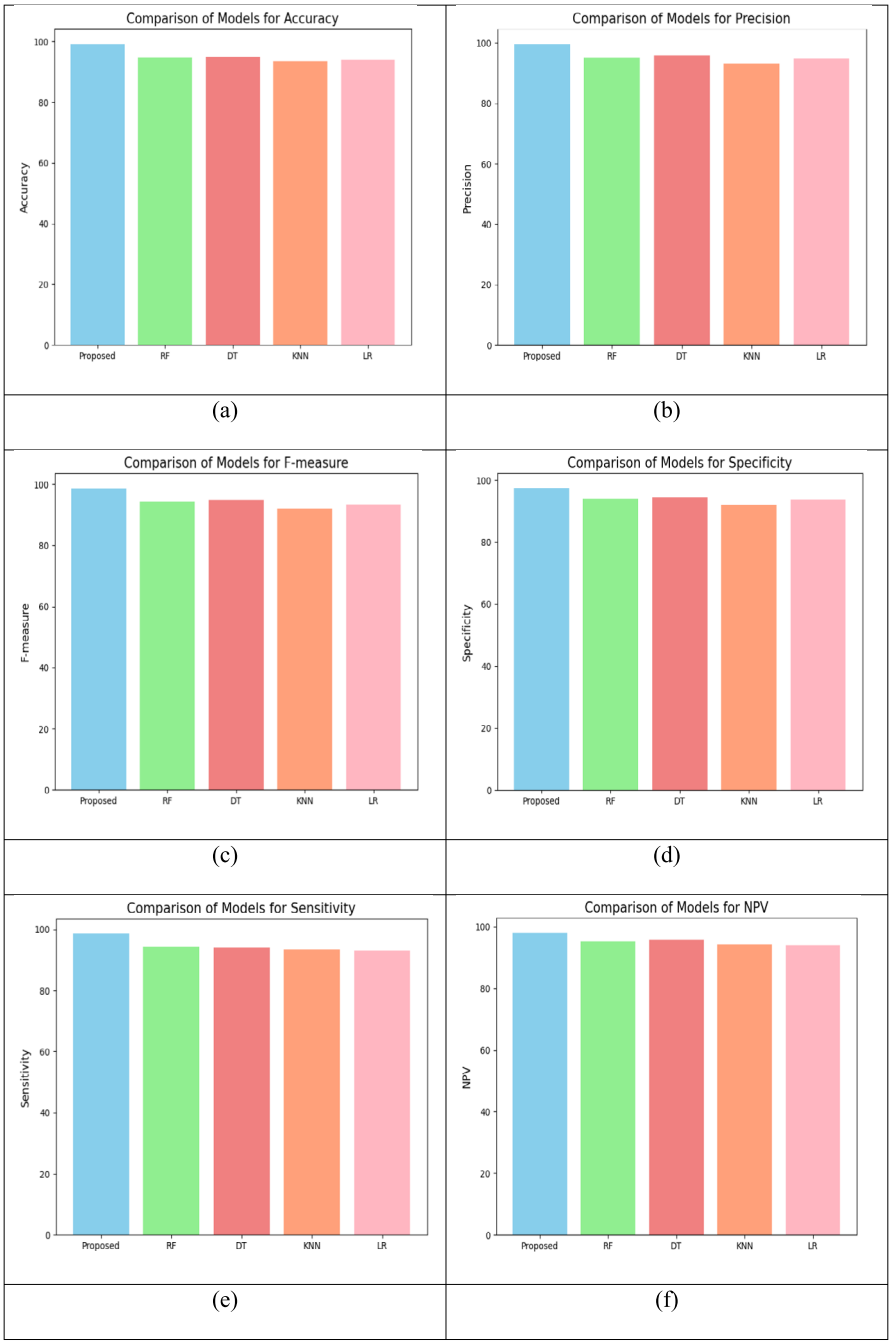


Fig. 3 Performance of Proposed TransGuard-QNet over Benchmark Models for (**a**) Accuracy, (**b**) Precision, (**c**) F1-Score, (**d**) Specificity, (**e**) Sensitivity, (**f**) NPV, (**g**) MCC, (**h**) FPR, and (**i**) FNR

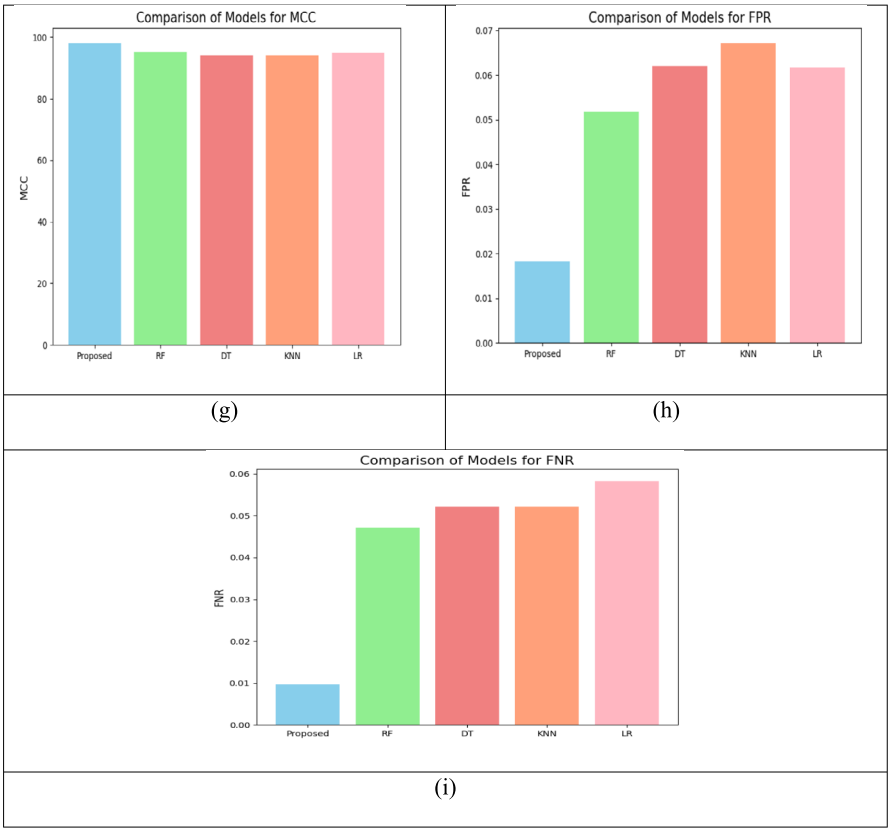


Fig. 3 (continued)

Table 3 Performance of Proposed TransGuard-QNet over Baseline Models for Various Performance Metrics

Model	Accuracy	Precision	F-measure	Specificity	Sensitivity	NPV	MCC	FPR	FNR
K-means	95.149	96.348	95.559	95.396	95.537	96.951	95.968	0.0446	0.0363
Xg Boost	96.818	96.064	95.539	95.553	95.325	96.473	95.667	0.0548	0.0449
QNN	94.545	95.544	93.348	93.914	94.931	95.553	94.787	0.0529	0.04189
QAE	95.379	95.089	94.896	94.983	93.994	94.603	95.082	0.0593	0.04409
Proposed Trans-Guard-QNet	**99.095**	**99.512**	**98.591**	**97.291**	**98.519**	**97.893**	**97.973**	**0.0182**	**0.00974**

Additionally, iXGBoost is parallel and distributed computing-enabled, thus making it highly capable of processing high-dimensional data and millions of instances, a widespread case in financial institutions. All these features cumulatively make the system highly responsive and reliable for real-time fraud detection and risk assessment purposes.

Table 4 Performance of proposed TransGuard-QNet over benchmark models for various performance metrics

Model	Accuracy	Precision	F-measure	Specificity	Sensitivity	NPV	MCC	FPR	FNR
K-means	94.621	95.081	94.087	94.025	94.418	95.21	95.108	0.0518	0.0472
XG-Boost	94.776	95.836	94.935	94.486	94.105	95.841	94.086	0.0621	0.0521
QNN	93.433	93.084	92.034	92.058	93.409	94.086	94.117	0.0672	0.0521
QAE	94.043	94.852	93.342	93.894	93.086	93.892	94.843	0.0618	0.05821
Proposd TransGuard-QNet	**99.095**	**99.512**	**98.591**	**97.291**	**98.519**	**97.893**	**97.973**	**0.0182**	**0.00974**

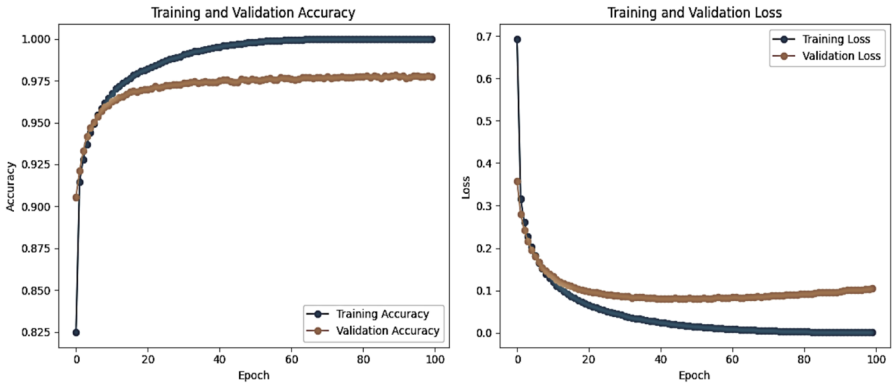


Fig. 4 Accuracy and loss performance of proposed TransGuard-QNet

5 Conclusion

This paper introduced TransGuard-QNet, an integrated hybrid QML framework that combined supervised and unsupervised models with LP for decision optimization for financial transaction risk assessment and prediction. Anomalies were found using QAEs and k-Means clustering techniques, and risk probability was estimated using iXGBoost and QNNs. The resulting predictions were combined to provide an outcome based on the optimized parameters, with a focus on achieving both high forecast accuracy and anomaly detection efficacy. The primary elements of feature engineering included statistics, behaviour, frequency, correlation characteristics, and mathematics. Additionally, the proposed model monitored outliers using iForest for enhanced robustness. According to LP, it also adjusted the weight criteria to identify final outputs. The proposed TransGuard-Qnet achieved better accuracy and outran other models. Future work will explore incorporating real-time data streams to enhance the model's adaptability to evolving fraud patterns. Advanced optimization techniques, such as quantum-inspired optimization, can further improve

decision-making efficiency. Expanding the model to support multi-language and cross-regional datasets will ensure broader applicability. Additionally, integrating explainable AI (XAI) methods will provide greater transparency and interpretability of risk predictions.

While quantum computing has encouraging potential for dealing with high-dimensional financial data processing and detecting subtle fraud patterns, its actual utilization in existing financial systems is restricted by hardware limitation, excessive costs of implementation, and the unavailability of mature development frameworks. Quantum models currently are experimental in nature and are not yet deployable for commercial use on a large scale. To address this, our research recommends implementing hybrid systems combining traditional machine learning models such as iXGBoost, Random Forests, GNNs with quantum-inspired optimization methods where possible. These hybrid systems can fill the gap by utilizing current infrastructure while setting the stage for quantum breakthroughs. Until quantum computing is more readily available, scalable classical models are the most practical and effective solution for real-time fraud detection in contemporary financial settings.

Funding On Behalf of all authors, the corresponding author states that they did not receive any funds for this project.

Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

References

Akash, T. R., Islam, M. S., & Sourav, M. S. A. (2024). Enhancing business security through fraud detection in financial transactions. *Global Journal of Engineering and Technology Advances, 21*(02), 079–087.

Ali, H., Zhang, Z., Safdar, R., Rasool, M. H., Yao, Y., Yao, L., & Gao, F. (2024). Fault detection using machine learning based dynamic ICA-distributed CCA: Application to industrial chemical process. *Digital Chemical Engineering, 11*, Article 100156.

Balcıoğlu, Y. S. (2024). Revolutionizing risk management AI and ML innovations in financial stability and fraud detection. In Navigating the Future of Finance in the Age of AI (pp. 109–138). IGI Global.

Bello, O. A., Ogundipe, A., Mohammed, D., Adebola, F., & Alonge, O. A. (2023). AI-driven approaches for real-time fraud detection in US financial transactions: Challenges and opportunities. *European Journal of Computer Science and Information Technology, 11*(6), 84–102.

Bhushan, S., Kumar, A., Pokhrel, R., Bakr, M. E., & Mekiso, G. T. (2024). Design based synthetic imputation methods for domain mean. *Scientific Reports, 14*(1), Article 4250.

Chatterjee, P., Das, D., & Rawat, D. B. (2024). Digital twin for credit card fraud detection: Opportunities, challenges, and fraud detection advancements. *Future Generation Computer Systems*. https://doi.org/10.1016/j.future.2024.04.057

Deepan, S., & Saravanan, M. (2024). Air quality index prediction using seasonal autoregressive integrated moving average transductive long short-term memory. *ETRI Journal*. https://doi.org/10.4218/etrij.2023-0283

Ding, L., Liu, L., Wang, Y., Shi, P., & Yu, J. (2024). An autoencoder enhanced light gradient boosting machine method for credit card fraud detection. *PeerJ Computer Science, 10*, Article e2323.

Dong, H., Liu, R., & Tham, A. W. (2024). Accuracy comparison between five machine learning algorithms for financial risk evaluation. *Journal of Risk and Financial Management, 17*(2), Article 50.

Du, Y., & Tao, D. (2024). On exploring the potential of quantum auto-encoder for learning quantum systems. IEEE Transactions on Neural Networks and Learning Systems.

Fatunmbi, T. O. (2024). Advanced frameworks for fraud detection leveraging quantum machine learning and data science in fintech ecosystems.

Ghaithan, A. M., Kondkari, M., Mohammed, A., & Attia, A. M. (2024). Optimal design of concentrated solar power-based hydrogen refueling station: Mixed integer linear programming approach. *International Journal of Hydrogen Energy, 86*, 703–718.

Goriparthi, R. G. (2023). AI-enhanced data mining techniques for large-scale financial fraud detection. *International Journal of Machine Learning Research in Cybersecurity and Artificial Intelligence, 14*(1), 674–699.

Guerand, J., & Imbert, C. (2023). Log-transform and the weak Harnack inequality for kinetic Fokker-Planck equations. *Journal of the Institute of Mathematics of Jussieu, 22*(6), 2749–2774.

He, W., Guo, X., Li, M., Zhang, M., & Chen, B. (2024). LPF/OGS: A low-pass filtering and overlapping group shrinkage denoising method for diesel engine fault diagnosis. *IEEE Sensors Journal*. https://doi.org/10.1109/JSEN.2024.3358061

Huang, Z., Zheng, H., Li, C., & Che, C. (2024). Application of machine learning-based k-means clustering for financial fraud detection. *Academic Journal of Science and Technology, 10*(1), 33–39.

Ijiga, O. M., Idoko, I. P., Ebiega, G. I., Olajide, F. I., Olatunde, T. I., & Ukaegbu, C. (2024). Harnessing adversarial machine learning for advanced threat detection: AI-driven strategies in cybersecurity risk assessment and fraud prevention.

Koo, K., Park, M., & Yoon, B. (2024). A suspicious financial transaction detection model using autoencoder and risk-based approach. *IEEE Access*. https://doi.org/10.1109/ACCESS.2024.3399824

Luo, S., Wang, B., Gao, Q., Wang, Y., & Pang, X. (2024). Stacking integration algorithm based on CNN-BiLSTM-Attention with XGBoost for short-term electricity load forecasting. *Energy Reports, 12*, 2676–2689.

Malik, P., Anand, A., Baliyan, A. K., Dongre, A., & Panwar, P. (2024). Credit risk assessment and fraud detection in financial transactions using machine learning. *Journal of Electrical Systems, 20*, 2061–2069.

Nemati, Z., Mohammadi, A., Bayat, A., & Mirzaei, A. (2024). Fraud risk prediction in financial statements through comparative analysis of Genetic Algorithm, Grey Wolf Optimization, and Particle Swarm Optimization. *Journal of Finance, 8*(1), 98–130.

Nur, R. (2025). Integrating quantum kernel methods with deep learning for real-time fraud detection in fintech platforms

Özbaltan, N. (2024). Applying machine learning to audit data: Enhancing fraud detection, risk assessment and audit efficiency. *EDPACS, the EDP audit, control, and security newsletter, 69*(9), 70–86.

Patil, D., Rane, N. L., Desai, P., & Rane, J. (2024). Machine learning and deep learning: Methods, techniques, applications, challenges, and future research opportunities. Trustworthy Artificial Intelligence in Industry and Society, 28–81.

Rahmad Ramadhan, L., & Anne Mudya, Y. (2024). A comparative study of Z-score and min-max normalization for rainfall classification in Pekanbaru. *Journal of Data Science, 2024*(04), 1–8.

Schatzki, L., Larocca, M., Nguyen, Q. T., Sauvage, F., & Cerezo, M. (2024). Theoretical guarantees for permutation-equivariant quantum neural networks. *Npj Quantum Information, 10*(1), Article 12.

Sorour, S. E., AlBarrak, K. M., Abohany, A. A., & Abd El-Mageed, A. A. (2024). Credit card fraud detection using the brown bear optimization algorithm. *Alexandria Engineering Journal, 104*, 171–192.

Tayebi, M., & El Kafhali, S. (2024). Performance analysis of metaheuristics based hyperparameters optimization for fraud transactions detection. *Evolutionary Intelligence, 17*(2), 921–939.

Udayakumar, R., Chowdary, P. B. K., Devi, T., & Sugumar, R. (2023). Integrated SVM-FFNN for fraud detection in banking financial transactions. *Journal of Internet Services and Information Security, 13*(3), 12–25.

Usman, A. U., Abdullahi, S. B., Liping, Y., Alghofaily, B., Almasoud, A. S., & Rehman, A. (2024). Financial fraud detection using value-at-risk with machine learning in skewed data. *IEEE Access*. https://doi.org/10.1109/ACCESS.2024.3393154

Vasudevan, S., Govindan, V., & Byeon, H. (2024). Online transaction fraud detection in the banking sector using machine learning techniques. *Edelweiss Applied Science and Technology, 8*(5), 864–872.

Wang, B., Chen, Y., & Li, Z. (2024a). A novel bayesian pay-as-you-drive insurance model with risk prediction and causal mapping. *Decision Analytics Journal, 13*, Article 100522.

Wang, B., Dong, Y., Yao, J., Qin, H., & Wang, J. (2024b). Exploring anomaly detection and risk assessment in financial markets using deep neural networks. *International Journal of Innovative Research in Computer Science & Technology, 12*(4), 92–98.

Yu, G., & Luo, Z. (2025). Financial fraud detection using a hybrid deep belief network and quantum optimization approach. *Discover Applied Sciences, 7*(5), Article 454.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

T. S. Sasikala[1] · S. Sivakami[1] · Prem Nisha G.[2] · R. Saranya[3]

✉ S. Sivakami
ssivakami411@gmail.com

T. S. Sasikala
sasikalaselva2012@gmail.com

Prem Nisha G.
nishainfo92@gmail.com

R. Saranya
saranyamvasan@gmail.com

1 School of Computing, Amrita Vishwa Vidyapeetham, Nagercoil, Tamil Nadu, India

2 Department of Computer science and Engineering, Arunachala College of Engineering for Women, Nagercoil, Tamil Nadu, India

3 Department of Artificial Intelligence and Data Science, Amrita Vishwa Vidhyapeetham, Nagercoil, Tamil Nadu, India